MALACALL · PAN-PLASMODIUM ANALYSIS

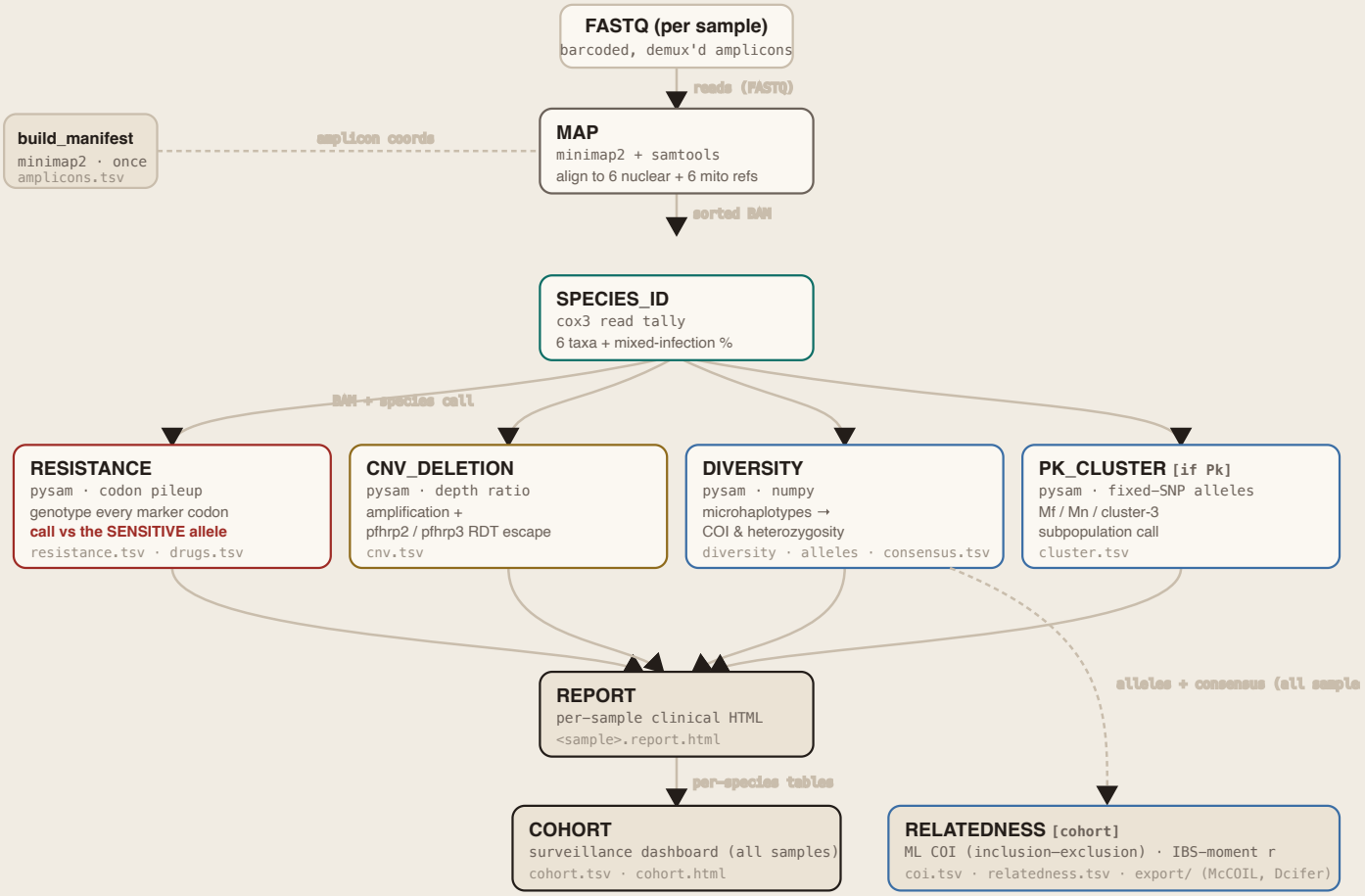
MalaCall — end-to-end analysis & reporting

One blood-spot FASTQ in; species ID (6 taxa, mixed infections), drug-resistance genotypes, RDT-escape deletions & gene amplification, parasite diversity, and *P. knowlesi* subpopulation out — attributed per species, then rolled up into a surveillance dashboard.

✓ validated end-to-end: 62 / 62 ground-truth assertions on simulated data

PIPELINE DAG

pre-process · species ID · resistance · diagnostic / CNV · diversity / cluster · reporting



Each sample flows MAP → SPECIES_ID; the species call then fans out to four independent, per-species analyses that reconverge into a clinical report, and all samples roll up into a cohort surveillance summary plus a cohort COI / relatedness analysis. Solid arrows carry data; the dashed arrows are the one-time amplicon manifest every stage consults and the per-sample diversity tables the relatedness stage pools across the whole cohort.

STAGES — TOOL, ACTION, OUTPUT

STAGE	TOOL	WHAT IT DOES	OUTPUT

DEMUX `[optional]`	`demux.py`	ONT custom-barcode runs only: split a pooled FASTQ by the panel's combinatorial 24-nt barcodes (CS1/CS2-anchored, both orientations, mismatch-tolerant), trimming to the amplicon. Native ONT barcodes are already demuxed by MinKNOW and skip this.	`per-sample fastq.gz · demux_stats.tsv`
build_manifest	`minimap2 + in-silico PCR`	One-time index: locate all 289 amplicons on the pan-reference (extract each amplicon, map for exact coordinates).	`amplicons.tsv / .bed`
MAP	`minimap2 (map-ont / sr) → samtools sort/index`	Align each sample's reads to the combined 6-nuclear + 6-mito reference; sorted, indexed BAM.	`<s>.bam`
SPECIES_ID	`pysam`	Tally mito **cox3** reads per taxon (counted at MAPQ 0 — cox3 is diagnostic by which contig it maps to). Proportions → present/absent; >1 present = mixed infection.	`<s>.species.tsv`
RESISTANCE	`pysam`	At every marker codon, build the codon from spanning reads, translate, and call **vs the sensitive allele** — so reference-is-mutant loci (pmdhfr 58/114) are correct. Aggregates to a per-drug interpretation.	`<s>.resistance.tsv · <s>.drugs.tsv`
CNV_DELETION	`pysam`	Read-depth of each CNV/deletion amplicon ÷ same-species single-copy controls → gene **amplification** (copy ratio) and **pfhrp2/pfhrp3 deletion** (RDT escape) when depth ≈ 0.	`<s>.cnv.tsv`
DIVERSITY	`pysam · numpy`	Per diversity amplicon: find variable positions → count microhaplotypes → per-locus alleles & He; within-host **COI** per species (90th-pct estimator). Also emits the per-locus **allele** and majority-**consensus** tables the cohort relatedness stage consumes.	`<s>.diversity.tsv · <s>.microhap.tsv · <s>.alleles.tsv · <s>.consensus.tsv`
PK_CLUSTER	`pysam`	Read the fixed cluster-diagnostic SNP alleles (chr1 C1A, chr6 C2J, chr8 cap380) → vote **Mf / Mn / cluster-3**. Runs only when Pk is present.	`<s>.cluster.tsv`
REPORT	`python`	Combine all per-sample tables into a clinical one-page HTML (species, per-species drug table, deletions/CNV, COI, cluster).	`<s>.report.html`
COHORT	`python`	Aggregate all samples → species distribution, per-drug resistance frequency, RDT-escape/amplification frequency, COI distribution.	`cohort.tsv · cohort.html`
RELATEDNESS `[cohort]`	`relatedness.py`	Cohort-level, dependency-free. **ML COI** per sample/species by the inclusion–exclusion probability that *m* clones cover the observed allele set given population frequencies (the categorical idea behind THE REAL McCOIL), regularised so the likelihood has an interior maximum. **Pairwise relatedness r** by a frequency-	`coi.tsv · relatedness.tsv · related_pairs.tsv · export/{realmccoil_*, dcifer_input}.tsv`

	adjusted IBS-moment on population-informative microhaplotypes (built from the cross-sample consensus). Also **exports** the McCOIL categorical matrix and Dcifer long table for publication-grade runs.

HOW TO RUN

MODE	COMMAND
Install the CLI	`pip install -e . → malacall --help`
Samplesheet from a run folder	`malacall samplesheet --platform ont|illumina|mgi --run-dir RUN --out samples.csv`
Run the pipeline	`malacall run --samplesheet samples.csv --outdir results --platform ont`
Single sample (no Nextflow)	`malacall sample --fastq S.fastq.gz --sample S --outdir results --platform ont`
Cohort roll-up	`malacall cohort results my_survey`
Cohort COI + relatedness	`malacall relatedness results my_survey`
Validate (simulated truth)	`python3 test/simulate.py … && python3 test/validate.py sim out → 62/62 PASS`

On `-profile localpy`: it only sets which local `python3` (the one with pysam/numpy/pandas) runs the stages — edit its path in `nextflow.config` to your own env, or use `--python <path>` or `-profile conda/docker`. `-profile captureforge` is a *retained alias* for the same thing; it is *not* the CaptureForge software (the analysis calls nothing from it) — that env was first built for the panel *design* work, where CaptureForge was actually used, and is simply reused here.

`MalaCall · pan-Plasmodium analysis pipeline v1.0 · Nextflow DSL2 + Python (pysam/numpy) · in-silico validated (62/62) · research pipeline, wet-lab pilot pending · full docs in analysis/MANUAL.md`